Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/02/2006
PI Name:	Powell, Robert L. Ph.D.		
Project Title:	STUDIES OF EMULSIONS DYNAMICS AND FLOWS USING MAGNETIC RESONANCE IMAGING		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	phone number chgd 6/2/06		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dungan, Stephanie (University of California Davis) Phillips, Ronald (University of California Davis)		
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	The overall goal of this work is to understand the stability of emulsion, especially as they relate to food syatems that might be used in low gravity applications. Since these will be high concentration, the best technique to assess this is using magnetic resonance imaging, MRI. Our first task has been to develop MRI procedures for measuring emulsion sizes dynamically, during flow. This has been accomplished. To undertake fundamental studies, techniques are being explored to make monodisperse emulsions so that changes in drop sizes can be monitored with the utmost accuracy. Because these techniques will likely involve the use of sample volumes that may be ten times smaller than used previously, a new MRI flow cell is being designed. As par tof our program to develop a wide specturm of systems used in our studies, we are also looking at the parameters that will allow us to tailor emulsion stability. The final part of this work is to develop models of flows of emulsions. A manuscript describing this work is under review. Ascertain criteria		

Task Description:	for the stability of emulsions in low gravity environments and the effect of flow on this stability. The group has developed procedures for determining flow fields (one and two-dimensional) and drop sizes for concentrated emulsions. This task combines those two procedures so that emulsion dynamics can be determined. Previous work has shown that emulsions can exhibit complicated flow phenomena due to slight differences in density between the dispersed and continuous phases. A second part of this work seeks to model these flows. There are no existing procedures for studying the dynamic drop size in flowing concentrated emulsions. This would permit a wide range of study with practial systems. By developing a flow model that can describe the flow behavior under well-defined conditions, it will be possible to model more complex flows of multiphase systems. For example, this work can be applied to determine the stabilty of processed foods in low gravity environments and also the stabilty of gas/liquid systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many systems of practical interest are emulsions, particularly food systems. The stability of these systems is crucial to their ultimate performance. This stability is influenced by imposed force fields, such as gravity or those arising from flow. By measuring the dynamic behavior of emulsions in flow those factors that will influence, for example, break-up and coalesence, and thereby also the stability can be determined. The modeling studies provide insight into flows of systems where small density differences may lead to large effects. In this sense, it is an analog of a low gravity environment on earth. This has begun to attract the attention of companies who hope to use these models for design and operation of high technology equipment.
Task Progress:	Over the last year, two graduate students (Vannarith Leang and Ian Klink), one post-doctoral fellow (Marcos D'Avila) and four Ph. D. s (Stephanie, Dungan, Ronald Phillips, Robert Powell and Jeffrey Walton) have been working on this project. In the last year the following has been accomplished: 1. New MRI pulse sequences for measuring drop dynamics under flow have been successfully tested. 2. A technique is now available for making narrow distribution emulsions. While this technique produces excellent results, it does not produce emulsions in quantities that are large enough to be used with the existing MRI flow cell. 3.A wide array of oil / water / surfactant systems have been tested. It is now possible to design systems that can be tailored to be stable under different conditions. 4. A manuscript on the modeling of flows of emulsions has been submitted along with new MRI data that can is used to verify the model. In the upcoming year, the following accomplishments are expected: 1. Articulation of first flow stability maps of concentrated emulsions 2. Design and construction of a new flow cell for use with our MRI system. This will allow the use of smaller sample volumes. 3. Experiments with narrow distribution emulsions. 4. Both graduate students working on this project with pass their Ph.D. Qualifying Exams. Also, based on the work just submitted for publication that developed a model for concentrated emulsions and verified the model with MRI experiments, a new collaboration has been stated with a major biotechnology company. They are interested in modeling flows inside a new generation of large scale fermentors. It appears that they will be funding a graduate student to work on this project. This will then allow the model to be further developed and used for the NASA funded research.
Bibliography Type:	Description: (Last Updated: 02/27/2004)